

DSAI Project

Debate Analysis: Convincing Argument Detection, AI-Generated Text
Detection, and Debate Generation with Fine-tuned LLMs

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1 Introduction

1.1 Project Background

This project was not among the topics initially proposed on Moodle. Our group wanted to explore and deepen our knowledge of NLP following the Machine Learning for Text Mining course taught by Matthieu Labeau. Since project selection was competitive, we took the initiative of contacting him directly to propose an open-ended project.

Our early ideas focused on empathy assessment in health-related discourse, pedagogical analysis of videos and articles, or greenwashing detection in ambiguous environmental communications. After discussing with our supervisor, we converged instead on the computational analysis of online debates.

This project uses the **ChangeMyView** (CMV) subreddit as its main data source, a platform where users share their opinions and award a Δ symbol to arguments that successfully changed their mind. This validation mechanism provides a natural label that is rare in web data, making CMV an ideal corpus for studying persuasion. From this framework, we structured the project into three axes, presented below.

1.2 Project Structure

The project follows a logical progression: understanding persuasion, detecting synthetic content, and then applying both components to arguments generated by fine-tuned LLMs.

The **first axis** focuses on the study of argument persuasiveness. Using the **Winning Argument Corpus** (WAC), a filtered and cleaned version of CMV, we aim to build a classifier capable of distinguishing a persuasive argument from a non-persuasive one. We explore several approaches: classic vector encoders (TF-IDF, Word2Vec, RoBERTa) followed by an SVM, a fine-tuned RoBERTa, and linguistic features inspired by the work of Tan et al. [1] capturing both the intrinsic style of the argument and its relationship with the original post. This classifier forms the core building block of the project, and will be reused in the third axis.

A natural complementary question arises: if language models can produce convincing arguments, can we tell them apart from human-written text? This is the challenge of the **second axis**, which addresses AI-generated text detection. Relying on three corpora of increasing difficulty, GriD, HC3 and M4GT-Bench, we evaluate several families of methods, from TF-IDF + SVM to full RoBERTa fine-tuning. The goal is to measure not only the raw performance of each approach, but also its ability to generalise across different generators and domains. This model will serve as an authenticity detector in the third axis.

The goal of the **third axis** is to fine-tune an open-source LLM (GPT-2, Qwen) on WAC data, then steer its generation to simultaneously maximise the persuasion score from Axis 1 and the authenticity estimated by Axis 2. Two strategies are explored: *prompt engineering* guided by the classifier coefficients, and *Best-of-N* sampling, where the best argument is chosen from several generated candidates. Due to hardware constraints, fine-tuning a 7-billion-parameter model on a consumer GPU led us to implement QLoRA to make this process tractable.

2 Axis 1: Modelling and Predicting Persuasion

In this section, we explore several ways to assess how persuasive an argument is. Our goal is to obtain the best possible classification model, capable of scoring and classifying an argument as persuasive or not. To this end, we use the Winning Argument Corpus (WAC), along with encoder-based methods such as W2V, TF-IDF or RoBERTa followed by an SVM or a fine-tuned linear layer, as well as less conventional approaches such as the pairwise method.

2.1 Winning Argument Corpus (WAC)

2.1.1 Dataset Presentation

The *Winning Argument Corpus* is built from the **r/ChangeMyView** subreddit, where an author presents an opinion (the *original post*, OP) and other users reply to try to change their mind. If the OP considers that a reply has changed their position, they award it a Δ (**delta**) symbol. Each discussion thus provides pairs of “winning” / “losing” replies to the same post, used to study what makes an argument persuasive. Below is an excerpt from a discussion taken from the dataset:

```
1 {"id": "t3_2cm3oh", "root": "t3_2cm3oh", "user": "nintendosixtyfoooour",
2   "reply-to": null, "text": "I go to the movies a lot and it seems to happen...",
3   "meta": {"success": null, "pair_ids": [], "score": null, ...}}
4
5 {"id": "t1_cjgsk1z", "root": "t3_2cm3oh", "user": "KevinWestern",
6   "reply-to": "t3_2cm3oh", "text": "Haven't you ever been near a sports fan...",
7   "meta": {"success": 1, "pair_ids": ["p_779"], "score": 757, ...}}
8
9 {"id": "t1_cjgv6jc", "root": "t3_2cm3oh", "user": "Notsuchameangirl",
10  "reply-to": "t3_2cm3oh", "text": "Clapping at the end of a movie you enjoyed...",
11  "meta": {"success": 0, "pair_ids": ["p_779"], "score": 9, ...}}
12
13 {"id": "t1_cjh1biu", "root": "t3_2cm3oh", "user": "TomasTTEngin",
14  "reply-to": "t3_2cm3oh", "text": "It's an effective way to remove popcorn residue...",
15  "meta": {"success": null, "pair_ids": [], "score": 26, ...}}
```

Figure 1: Excerpt from `utterances.jsonl` (a WAC file)

In `utterances.jsonl`, `meta.success` indicates the status of each comment relative to the OP: 1 marks a comment that received a delta (**persuasive**) and was included in an argument pair, and 0 marks the other member of that pair, which did not (**non-persuasive**) and serves as a comparison point. The `null` label is by far the most frequent, covering any comment not retained in a pair, either because it is not a genuine persuasion attempt or because it cannot be meaningfully compared to the pair’s argument. The link between the two members of a pair is given by `meta.pair_ids`. `meta.score` represents a comment’s upvote count; we do not use it here.

2.1.2 Simplification of Axis 1 & Dataset Cleaning

The original goal of Axis 1 was to detect the Δ among all comments in a post. After considering several approaches, we decided to significantly simplify the problem by framing it as a binary classification task: for each original post, we only keep the 0/1 argument pairs, discarding all

comments labelled `null`. We can then build a model capable of distinguishing a winning comment from a losing one.

This simplification is nevertheless quite rough and discards a significant amount of information. The entire tree structure of the debate disappears, along with the chains of argumentation and rebuttal that sometimes lead to a Δ , and the temporal position of each comment. The upvote score of each comment is also ignored, even though it could add information, distinct from the OP’s persuasion.

```

1 id,root,op_text,text,success,pair_id
2 t1_cjgskiz,t3_2cm3oh,"I go to the movies a lot and it seems to happen...",
3 "Haven't you ever been near a sports fan...",1.0,p_779
4 t1_cjgv6jc,t3_2cm3oh,"I go to the movies a lot and it seems to happen...",
5 "Clapping at the end of a movie you enjoyed...",0.0,p_779

```

Figure 2: Excerpt from `WAC.csv` , the two rows forming pair `p_779`

Here is an excerpt from our filtered dataset in CSV format, with 6 columns: `id` (comment identifier), `root` (identifier of the original post being replied to), `op_text` (the original post), `text` (the comment), `success` (winning or losing label) and `pair_id` (identifier linking the two comments of the same pair). In total, we obtain 3866 winning/losing argument pairs, giving a dataset of **7732** perfectly balanced arguments, which justifies our use of standard Accuracy to evaluate our models going forward.

2.2 Approach Using Classic Vector Representations

2.2.1 Encoders : TF-IDF, Word2Vec & RoBERTa

Our first approach vectorises comments using different classic encoders, then trains an SVM on the resulting vectors. We compare the representations produced by these encoders to assess how much an encoder’s semantic richness influences the SVM’s ability to distinguish a persuasive comment from a non-persuasive one. The three encoders tested are:

- **TF-IDF**: A representation based on word frequency, which captures discriminating vocabulary without taking word order or context into account.
- **Word2Vec**: A model that learns a dense vector for each word from the words that frequently surround it, so words with similar meanings obtain close vectors in the vector space.
- **RoBERTa**: A pre-trained Transformer model that produces a context-dependent vector for each word, capturing nuances of meaning that static approaches like Word2Vec cannot.

For the last two encoders, the final vector for a comment is the average of the representations of each word it contains.

2.2.2 Results and Discussion

We test two strategies. The first vectorises only the comment, ignoring the original post (OP), to evaluate the intrinsic quality of the argument independently of context. The second concatenates the OP and the comment before vectorising, adding debate context, since the same argument may be more or less relevant depending on the OP’s initial position. In both cases, training uses 80% of the dataset, with 20% reserved for testing. The results are as follows:

Model	Text only	Text + OP
Word2Vec + SVM	0.56	0.49
TF-IDF + SVM	0.51	0.32
RoBERTa + SVM	0.54	0.35

Table 1: SVM Accuracy by encoder and input context

All results stay close to random (0.50) regardless of the encoder, confirming the inherent difficulty of the task: distinguishing a persuasive argument from a non-persuasive one purely from textual content is a fundamentally hard problem, even for rich semantic representations.

In the text-only setting, Word2Vec gets the best accuracy (0.56), marginally ahead of RoBERTa (0.54) and clearly ahead of TF-IDF (0.51). This is somewhat counterintuitive, since RoBERTa’s contextual representations should be richer, but averaging over all tokens collapses them into a document vector that loses much of that context. Word2Vec’s averaged embeddings suffer the same aggregation loss, but their denser geometry seems more suited to linear SVM separation on a small dataset.

The most striking finding is the accuracy drop once the OP is concatenated to the comment. For TF-IDF, accuracy collapses from 0.51 to 0.32, well below chance, with a similar effect for RoBERTa (0.54 to 0.35). This dilution effect comes from the pairwise structure of the dataset: since both comments in a pair respond to the same OP, concatenating it artificially brings their vectors closer together, cancelling the discriminative signal. These results show that naive context injection via concatenation is counter-productive, motivating the more principled approaches in Sections 2.3 and 2.5.

2.3 Intrinsic Features and Interplay

2.3.1 Paper’s features

These results show that vectorising the comment alone, or naively concatenating the OP and the comment, doesn’t fully exploit the relationship between the two texts. We therefore turn to Tan et al. [1], who worked on the same **ChangeMyView** corpus and proposed around forty linguistic features organised into two families.

The first family, **interplay**, captures the relationship between the comment and the original post: for each pair (comment A , original post O), they compute the number of shared words $|A \cap O|$, the reply-side fraction $\frac{|A \cap O|}{|A|}$, the OP-side fraction $\frac{|A \cap O|}{|O|}$, and the Jaccard similarity $\frac{|A \cap O|}{|A \cup O|}$, applied to stopwords, content words and the full vocabulary. The key finding is that

persuasive arguments are less similar to the OP in content words, while staying more similar in stopwords: they bring new ideas while aligning stylistically with their interlocutor.

The second family groups **intrinsic or stylistic** descriptors of the comment: length (#words, #sentences, #paragraphs), pronoun usage (first-person singular and plural, second-person), presence of hyperlinks, example markers, hedges, and Markdown formatting (italics, bold, lists).

2.3.2 Results and Discussion

Class	Precision	Recall	F1-score	Support
0 (Losing)	0.58	0.70	0.63	774
1 (Winning)	0.62	0.50	0.54	773

Table 2: Performance of Paper’s Features + SVM; **Accuracy = 0.59**

Tan et al.’s linguistic features reach an accuracy of 0.59, a clear improvement over the encoder-based approaches from Section 2.2. This is notable given how compact the feature vector is: a few hand-crafted features outperform dense pre-trained representations, suggesting that persuasiveness here is better captured by structured rhetorical signals than by raw content.

The class-level breakdown reveals an asymmetry: the model recalls losing arguments more reliably (0.70) than winning ones (0.50). It is thus more confident at spotting what makes an argument fail than what makes it succeed. Interplay and stylistic markers can flag obvious weaknesses, but the positive signals of persuasion stay subtler and harder to capture with a fixed feature set. The SHAP analysis in Section 2.3.3 looks at which features actually drive these decisions.

2.3.3 Feature Analysis with SHAP

To interpret the SVM model, we use the SHAP method. At the local level, it quantifies each variable’s impact on a prediction: a positive value pushes towards the “persuasive” class (1), and vice versa. At the global level, aggregating these values over the test set gives the overall feature importance ranking.

Interplay features dominate by a wide margin, confirming Tan et al.’s findings. One apparent contradiction stands out: a high `op_frac_all` is associated with positive SHAP values, while a high `jaccard_all` generates negative ones. This comes down to normalisation: `op_frac_all` = $\frac{|A \cap O|}{|O|}$ measures OP coverage, whereas Jaccard = $\frac{|A \cap O|}{|A \cup O|}$ penalises longer comments. A persuasive argument therefore covers a large part of the OP’s vocabulary while introducing a lot of new vocabulary, low Jaccard, high `op_frac`, consistent with Tan et al. A high `reply_frac_all` generates negative SHAP values: paraphrasing the OP without adding anything new hurts persuasiveness. Among intrinsic features, `frac_links` and `n_sentences` contribute positively, while formatting (`n_italics`) and lexical diversity (`word_entropy`) have only a marginal impact.

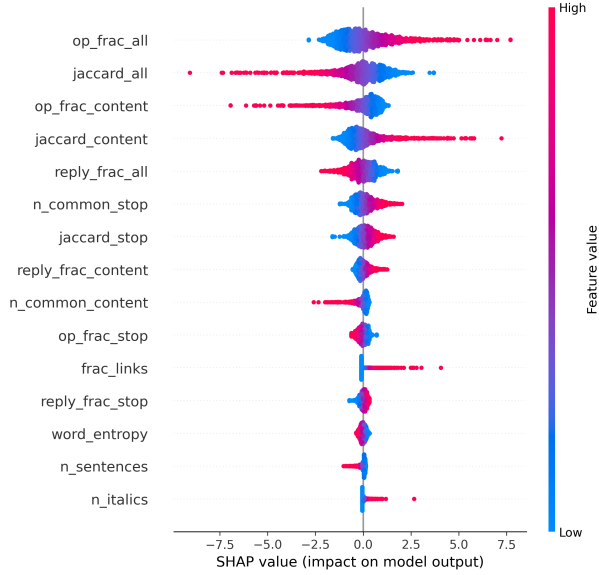


Figure 3: Feature importance representation

2.4 The Pairwise Method

This section complements our main approach. We focus here on the pairwise method, as implemented in Tan et al. Rather than classifying each argument independently, we feed the classifier directly with winning/losing argument pairs, and the task consists in identifying which of the two is persuasive. This reformulation simplifies the problem, since both arguments respond to the same OP and are therefore directly comparable, which leads us to expect better performance.

In practice, we encode each argument in the pair with our four vectorisation methods (TF-IDF, Word2Vec, RoBERTa and Tan et al.’s features), then subtract the two vectors to obtain a single vector representing the contrast between the two arguments. This vector is then fed into a Logistic Regression classifier to predict which argument is more persuasive.

Encoder	Classic Approach (SVM)	Pairwise Method (LogReg)
TF-IDF	0.51	0.58
W2V	0.56	0.59
RoBERTa	0.54	0.51
Paper’s Features	0.59	0.63

Table 3: Accuracy comparison between SVM and Pairwise approaches

The pairwise reformulation yields consistent gains across all encoders. The improvement is most pronounced for TF-IDF (0.51 to 0.58) and Paper’s Features (0.59 to 0.63), while RoBERTa is the only encoder that does not benefit from the reformulation (0.54 to 0.51). This exception is likely due to the vector subtraction step: RoBERTa’s contextual embeddings are not designed to be compared by difference. Overall, these results confirm that explicitly comparing argument pairs, rather than classifying each independently is a more natural framing for the task, consistent with the pairwise structure of the WAC dataset itself.

2.5 Fine-tuned RoBERTa

The results so far highlight a clear limitation: vectorising arguments, even with a powerful encoder like RoBERTa, is not enough to capture the dynamic relationship between an argument and the original post. We therefore moved to a full fine-tuning of RoBERTa, adapting all model weights to our pairwise classification task.

2.5.1 The Multiple Choice Architecture

We keep the pairwise approach from Section 2.4, but implementing it with a Transformer raises a concrete problem: the strict 512-token limit. Naively concatenating the OP and both arguments would force heavy truncation, losing crucial information such as the argument’s conclusion, often where persuasion is decided.

We therefore reformulated the task using the `RobertaForMultipleChoice` architecture. The model separately evaluates the pair (OP, `Argument 0`), then the pair (OP, `Argument 1`). A final linear layer then compares the two representations to predict the winner. This allocates 512 tokens to each pair, twice the budget of a naive three-text concatenation, while letting the model capture the OP–argument relationship via the Transformer’s internal cross-attention.

2.5.2 Data Preparation and Bias Handling

Training this model correctly required adapting our dataset and taking a few precautions.

The first concerns **position bias**. We randomly shuffled the position of the winning argument for each pair. Without this, the network would quickly learn to systematically predict the same position to minimise its error, without ever reading the text.

The second relates to **data leakage**. The train/validation/test split was performed by grouping by post identifier, ensuring a test argument always responds to an OP the model has never seen during training.

2.5.3 Results and Interpretation

The final evaluation was conducted on a test set of 387 pairs never seen during training. The detailed results are presented in Table 4.

Class	Precision	Recall	F1-score	Support
0 (Option 0 wins)	0,69	0,76	0,72	187
1 (Option 1 wins)	0,75	0,69	0,72	200

Table 4: Performance of the fine-tuned RoBERTa Cross-Encoder trained on 4 epoch; **Accuracy = 0,72**

The fine-tuned RoBERTa Cross-Encoder reaches 72% accuracy, a substantial improvement over all previous approaches. Both classes get a balanced F1-score of 0.72, confirming the model doesn’t show the recall asymmetry seen with hand-crafted features in Section 2.3.

This validates the Cross-Encoder architecture: encoding each (OP, argument) pair separately within a 512-token budget lets the model attend to the post–candidate relationship without suffering the dilution effect from Section 2.2. Full fine-tuning thus lets RoBERTa learn persuasion-

specific patterns directly from raw text, replacing Tan et al.’s features while clearly surpassing them (0.72 vs. 0.63). This model is our strongest scoring function and will be used as such in the Axis 3 generation pipeline.

Model	Approach	Accuracy
TF-IDF + SVM	Classic	0,51
Word2Vec + SVM	Classic	0,56
RoBERTa + SVM	Classic	0,54
Paper’s Features + SVM	Classic	0,59
TF-IDF	Pairwise	0,58
Word2Vec	Pairwise	0,59
Paper’s Features	Pairwise	0,63
Fine-tuned RoBERTa	Cross-Encoder	0,72

Table 5: Summary of all Axis 1 approaches on the WAC corpus

3 Axis 2: Automated Detection (Human vs. AI)

Axis 2 addresses a binary safety classification problem: determining whether a text was written by a human or generated by an artificial intelligence model. In the overall project architecture, this module acts as an authenticity filter: it assesses whether the arguments produced or analysed retain a sufficiently human-like signature.

Rather than presenting each experiment separately, we group the results here according to a comparative logic: the same encoder families, the same linear classifiers where possible, but applied to three datasets of increasing difficulty. This organisation makes the strengths and limitations of each approach easier to see.

3.1 Data Corpora: HC3, GriD and M4GT

The evaluation relies on three complementary corpora, chosen to test both raw performance and the generalisation ability of the detectors:

- **HC3 (Human ChatGPT Comparison Corpus):** A corpus built around pairs of human and ChatGPT responses. It provides a useful framework for measuring the separability between human style and the style of a conversational model.
- **GriD:** A dataset used as an intermediate benchmark, allowing comparison of lexical, semantic and contextual representations on a smaller number of examples.
- **M4GT-Bench:** A more heterogeneous corpus, integrating multiple generators and multiple domains. It is therefore more representative of an Out-Of-Distribution scenario, where the detector must not only recognise a specific model, but generalise to varied generation styles.

3.2 Methods Evaluated

We compared four families of approaches, ranging from simple statistical representations to full Transformer fine-tuning:

3.2.1 TF-IDF + SVM

The TF-IDF method transforms each text into a high-dimensional sparse vector, where each component represents the relative importance of a term in the document and in the corpus. This representation is then fed to a linear SVM. Its main advantage is its simplicity, speed and robustness on large text corpora. It captures differences in vocabulary, phrasing and lexical frequency between human and generated texts very well.

3.2.2 Word2Vec + SVM

Word2Vec projects words into a dense space where geometric proximities reflect semantic similarities. To represent a full document, word vectors are aggregated and then passed to a linear SVM. This method goes beyond simple lexical frequency, but loses a significant part of syntactic information and word order.

3.2.3 SentenceTransformer / RoBERTa + SVM

Models such as SentenceTransformer or RoBERTa produce contextual embeddings: the representation of a word or sentence depends on the global context. These vectors are then used as input to an SVM. This approach aims to capture finer signals than TF-IDF or Word2Vec, but its performance depends heavily on the quality of the embedding and its adaptation to the domain.

3.2.4 RoBERTa Fine-tuning

Finally, we tested a full fine-tuning of `roberta-base` on GriD using `AutoModelForSequenceClassification`. Unlike RoBERTa + SVM, the model is not only used as a feature extractor: its weights are adjusted to directly optimise human/AI classification. This approach is more costly, but can capture contextual and stylistic signals much more specific to the dataset.

3.3 Comparative Experimental Results

The table below summarises all the results extracted from the classification reports. To keep the overview compact, we report the overall accuracy, per-class F1-scores and total support. Detailed precision and recall values are then discussed in the per-dataset analysis.

Table 6: Global comparison of AI detection methods on GriD, HC3 and M4GT.

Dataset	Method	Accuracy	F1 class 0	F1 class 1	Support 0	Support 1
GriD	TF-IDF + SVM	0,93	0,96	0,81	1053	250
GriD	Word2Vec + SVM	0,91	0,94	0,73	1053	250
GriD	RoBERTa fine-tuning	0,99	1,00	0,98	1076	227
HC3	TF-IDF + SVM	0,98	0,99	0,97	10268	3332
HC3	Word2Vec + SVM	1,00	1,00	1,00	10268	3332
HC3	RoBERTa + SVM	0,95	0,97	0,91	10268	3332
M4GT	TF-IDF + SVM	0,88	0,85	0,89	1667	2333
M4GT	Word2Vec + SVM	0,76	0,68	0,81	1667	2333
M4GT	RoBERTa + SVM	0,83	0,80	0,85	1667	2333
M4GT	RoBERTa fine-tuning	0,98	0,98	0,98	1667	2333

3.4 Latent Space Visualisation: t-SNE and UMAP

To complement the metrics, we projected the high-dimensional representations into two dimensions. For the HC3 dataset, we chose **t-SNE** (t-Distributed Stochastic Neighbor Embedding) over standard PCA. PCA is a linear projection that attempts to preserve global variance, often resulting in heavily overlapping clusters for complex text embeddings. Conversely, t-SNE excels at preserving local neighbourhoods and non-linear relationships.

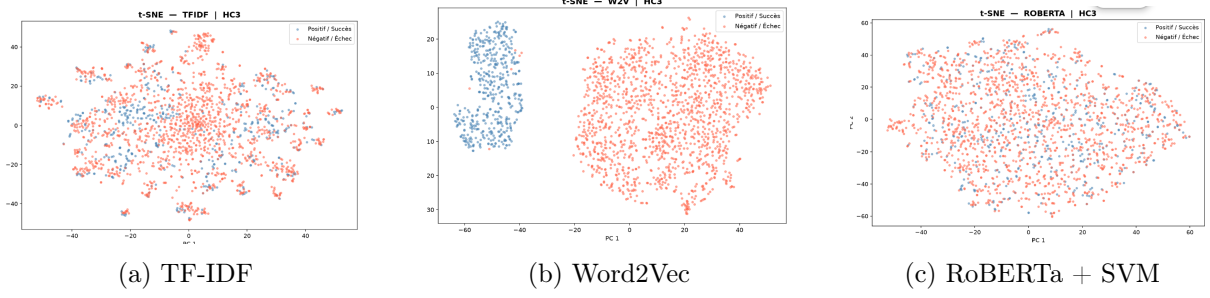


Figure 4: t-SNE projections on HC3.

On the HC3 t-SNE projections, we observe that TF-IDF and RoBERTa+SVM are very mixed between the positive and negative classes. In contrast, Word2Vec is perfectly separated, which aligns with its near-perfect accuracy on this dataset.

To analyse the much more challenging M4GT dataset, we transitioned to **UMAP** (Uniform Manifold Approximation and Projection). We chose UMAP here because it better preserves global topological structure and scales more effectively to the high complexity of a multi-generator corpus. We deliberately focused our UMAP analysis solely on M4GT; its high diversity and out-of-distribution characteristics make its latent structure far more challenging and scientifically interesting to explore than the HC3 dataset.

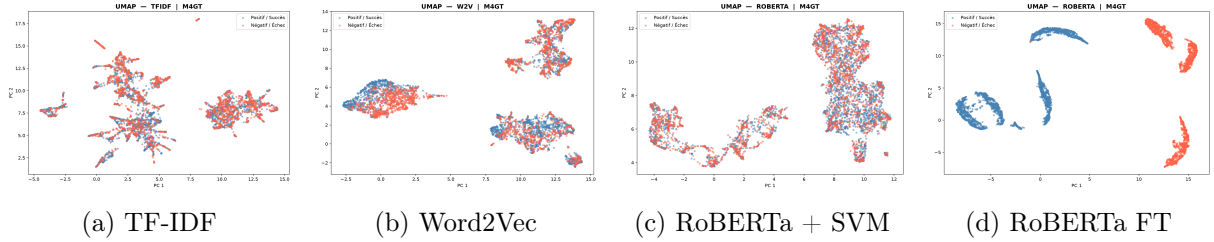


Figure 5: UMAP projections on M4GT.

The UMAP projections for M4GT reveal that TF-IDF and RoBERTa+SVM do not separate the classes any better, remaining heavily mixed. Word2Vec separates the classes noticeably worse than it did on HC3, although it still maintains slightly better clustering than the other frozen encoders. Crucially, the fine-tuned RoBERTa model achieves a perfect separation. This visual confirms that fine-tuning fundamentally reorganises the latent space: instead of clustering texts by semantic meaning, the network learns a manifold strictly dedicated to isolating the subtle structural artifacts of AI generation.

3.5 Per-Dataset Analysis

3.5.1 GriD: advantage to supervised fine-tuning

On GriD, the TF-IDF + SVM baseline already achieves 93% accuracy, with an F1-score of 0.96 for class 0 and 0.81 for class 1. The MiniLM + SVM model drops slightly to 91% accuracy, mainly due to lower recall on the minority class. RoBERTa fine-tuning clearly dominates with 99% accuracy and F1-scores of 1.00 and 0.98. This shows that, on a controlled-size corpus, supervised adjustment of a Transformer can capture highly discriminating contextual signatures.

3.5.2 HC3: Strong separability and the ChatGPT signature

HC3 is the dataset on which scores are highest. TF-IDF + SVM achieves 98% accuracy, while Word2Vec + SVM achieves an accuracy rounded to 100%. While these results seem exceptional, a critical analysis of the corpus explains this phenomenon. HC3 was built using early versions of ChatGPT (GPT-3.5). This model exhibits a highly formatted, recognisable, and homogeneous "robotic" style (frequent use of bullet points, polite disclaimers, and lack of emotional variability).

Consequently, the lexical distribution of ChatGPT in HC3 is extremely distinct from human internet users. Even a basic lexical method like TF-IDF easily picks up on these repetitive patterns. This near-perfect score is therefore more indicative of the dataset's simplicity and the stereotypical nature of early ChatGPT, rather than the intrinsic robustness of our detectors.

3.5.3 M4GT: The Out-Of-Distribution generalisation challenge

M4GT is the most challenging corpus. For frozen encoders, the performance drop is significant: TF-IDF + SVM drops to 88% accuracy, RoBERTa + SVM to 83%, and Word2Vec + SVM falls to 76%.

Unlike HC3, M4GT is a multi-generator corpus (Llama, Mistral, Cohere, etc.) covering multiple domains. The "AI signature" is no longer a single homogeneous style. Word2Vec struggles severely because modern LLMs have learned to use a vocabulary that perfectly mimics human semantic distribution. TF-IDF resists slightly better because it captures broader n-gram frequencies. This highlights the core vulnerability of static AI detection: detectors overfitting to a specific model's quirks fail to generalise to the broader, out-of-distribution (OOD) landscape.

However, **RoBERTa fine-tuning completely reverses this trend**, achieving the best result by a wide margin (around 98% accuracy). While static embeddings fail to capture the subtle, diverse structural artifacts of multiple generators, full fine-tuning allows RoBERTa to adapt its internal attention heads specifically to the stylometric nuances of AI generation across different models. It transcends mere vocabulary or semantic similarity to detect deep syntactic and structural regularities inherent to LLM autoregressive generation, making it vastly superior for OOD detection.

3.6 Summary and Pipeline Choice

These experiments show that AI detection does not reduce to model complexity. TF-IDF remains extremely competitive despite its simplicity — it even outperforms dense embeddings on M4GT. Word2Vec performs well on HC3 but generalises poorly to M4GT. Fine-tuned RoBERTa excels on GriD, while RoBERTa used as a frozen encoder followed by an SVM does not systematically dominate. For the full pipeline, this justifies using fine-tuned RoBERTa as an authenticity judge when available, and TF-IDF as a robust low-cost baseline. The key methodological takeaway is that strong performance on a homogeneous dataset like HC3 does not guarantee reliable generalisation on a more open corpus like M4GT. A theoretical conclusion also emerges regarding frozen deep models. RoBERTa + SVM yielded mixed results (83% on M4GT, behind TF-IDF) because pre-trained RoBERTa embeddings are optimised for semantic similarity, not stylometry. A human and an AI text expressing the same idea will cluster closely in a frozen semantic space,

confusing a linear SVM. Full fine-tuning (98% on M4GT), by contrast, reorganises the embedding space and trains the model to focus on structural authenticity markers — which justifies always preferring a fine-tuned Transformer over a static encoder + SVM pipeline when resources allow.

4 Axis 3: Strategic Generation and LLM Fine-tuning

Axis 3 represents the convergence of our previous efforts. The goal is twofold: generate arguments capable of shifting human opinion (Axis 1). Then with our detection algorithms (Axis 2) we could detect if the argument looks AI-generated or not. However it's important to recall that this part is very experimental and we didn't have time to discuss a lot on this axis with our supervisors. Indeed, some results that we will find can have a big score but since our 2 previous models (especially the models of axis 1) are not perfect, the results can be biased.

4.1 Fine-Tuning and QLoRA Optimization

To generate convincing and relevant arguments, using a pre-trained large language model (LLM) (such as Qwen or GPT-2) is not always enough, the model also needs to adapt to the rhetorical style and specific format of our data from the *Change My View* (WAC) dataset. To do this, we fine-tuned the model so it would learn to predict only "winning" arguments (the ones that managed to convince the original author).

However, fully fine-tuning models with several billion parameters needs huge computational resources. To solve this problem, we used the **QLoRA** (Quantized Low-Rank Adaptation) approach, which is a PEFT (Parameter-Efficient Fine-Tuning) optimization method.

4.2 Generation Strategy: Best of N

To improve the relevance of our answers, we use the **Best of N** decoding strategy (also called *Rejection Sampling*).

This method separates generation from evaluation, so it avoids the limits of simple greedy generation. The process relies on the models from our different axes and happens in three steps:

1. **Candidate generation (LLMs):** For a given request, the system generates N different answers. We use **GPT-2** (124M parameters) for our initial tests, as well as **Qwen2.5-3B-Instruct** (3 billion parameters). Qwen was fine-tuned with QLoRA (4-bit quantization and LoRA), which allows efficient learning without needing too much video memory.
2. **Selection by Persuasion (Axis 1):** The N generated candidates are then evaluated by our Axis 1 model. This is a **RoBERTa-base** (125M parameters) fine-tuned with a **pairwise** approach (`RobertaForMultipleChoice`). This model compares the arguments and gives a persuasion score to each one. The candidate with the best score becomes our final answer.
3. **Authenticity Evaluation (Axis 2):** Axis 2 does not influence the final choice, it works as an a posteriori check instead. We pass the selected answer through another **RoBERTa** model, this time fine-tuned specifically on the **M4GT** dataset (made for AI detection). This lets us measure if our most persuasive argument keeps a natural style or if it risks being detected as machine-generated.

4.3 Results

4.3.1 Statistics of persuasion and authenticity scores

Statistic	persuasion_score	persuasion_acc	IA_score	IA_acc
mean	1.71	0.85	-2.37	0.09
std	1.35	0.18	1.67	0.13
min	-0.88	0.29	-4.99	0.01
25%	0.70	0.67	-3.86	0.02
50%	1.79	0.86	-2.65	0.07
75%	2.93	0.95	-1.26	0.22
max	4.01	0.98	2.59	0.93

Table 7: Statistics of persuasion and authenticity scores on 50 arguments generated by our fine-tuned **GPT-2**

Statistic	persuasion_score	persuasion_acc	IA_score	IA_acc
mean	1.27	0.78	0.81	0.69
std	1.39	0.24	0.97	0.21
min	-2.35	0.09	-1.28	0.22
25%	0.55	0.63	0.27	0.57
50%	1.36	0.80	0.91	0.71
75%	2.26	0.91	1.32	0.79
max	3.94	0.98	3.39	0.97

Table 8: Statistics of persuasion and authenticity scores on 50 arguments generated by our fine-tuned **Qwen 2.5**

4.3.2 Concrete examples of generated arguments

OP (Message original)	Reponse de l'IA (LLM)	Axe 1 (% Convaincant)	Axe 2 (% Généré par IA)
If you find yourself uninformed about representatives or issues, you should not nor should you be encouraged to vote. This to me seems common sense, yet in many or most elections people are attempting to get people to actively vote who otherwise would not, such as teachers giving extra credit for voting or businesses offering incentives to employees who vote. I understand people should be encouraged to understand the issue at hand and to research representatives, however «people who do not should not be encouraged to vote.» The reasoning behind this is simply that these choices should not be made at a spur of the moment or without conviction, as then we get representatives voted in who hold no qualifications but who the voter may have "heard of once or twice." Thank you for your time and I am honestly willing to change my view.	UPDATE (7/8) http://www-onlinejournalmagazinenownewsbeatingshotlineroompress.com/#post_154350&page=3 [REF]	55.2%	5.5%

Figure 6: GPT-2 - Example 1

OP (Message original)	Reponse de l'IA (LLM)	Axe 1 (% Convaincant)	Axe 2 (% Généré par IA)
If you find yourself uninformed about representatives or issues, you should not nor should you be encouraged to vote. This to me seems common sense, yet in many or most elections people are attempting to get people to actively vote who otherwise would not, such as teachers giving extra credit for voting or businesses offering incentives to employees who vote. I understand people should be encouraged to understand the issue at hand and to research representatives, however «people who do not should not be encouraged to vote.» The reasoning behind this is simply that these choices should not be made at a spur of the moment or without conviction, as then we get representatives voted in who hold no qualifications but who the voter may have "heard of once or twice." Thank you for your time and I am honestly willing to change my view.	1) People don't necessarily know what they're doing when it comes to politics. A lot of folks are perfectly capable of making informed decisions, but some aren't. You might want to encourage them to try harder, but you can't force them to do so. This means that there's going to be an awful lot of crap being voted on by people who wouldn't necessarily vote if they had their heads up their asses. 2) It's hard to tell whether someone has thought things through before casting their vote. We could use better tools than just looking at their name on the ballot to make sure that they've done enough reading and thinking to back up any claims they make about themselves. But those tools don't exist right now (or soon will). So we're left with asking people whether they've given the matter much thought – and the honest answer to that question is almost always "no". I'm not saying that we should let anyone off the hook entirely, mind you. There are definitely plenty of cases where people can't possibly have considered an important issue, because they didn't have the chance to. Those are still valid reasons to abstain from voting, obviously. But even here, we're relying on the person to self-report accurately – which, again,	48.0%	78.0%

Figure 7: Qwen2 - Example 1

OP (Message original)	Reponse de l'IA (LLM)	Axe 1 (% Convaincant)	Axe 2 (% Généré par IA)
Where I went to school (Australia) we are taught that there are 7 continents which are as follows: North America, South America, Europe, Asia, Africa, Australia, and Antarctica. I believe a bit of common sense will tell you that there is in fact, 6, which would come about by combining Europe and Asia as there is an acute lack of an ocean dividing them, and should therefore be considered one and the same; Eurasia. Most people will tell you that they are divided because of such a huge cultural divide, but I think that that argument is flawed in that many completely different cultures exist on the same continent (e.g. China and the Middle East). So if we were to choose the number of continents based on culture there would be an endless amount of completely subjective possibilities, which means that this method is irrelevant. One might argue that if we were to ignore culture and combine Europe and Asia, we would also have to include Africa to create a super-continent, and we would have to combine North and South America to make the Americas (for arguments sake, let's forget about the Suez and Panama canals). However, I would argue that this is different as the land bridges between them are vastly smaller. So, change my view; it makes much more sense to have 6 continents rather than 7.	「= how large or small [I like] those differences since none could ever add up[/PRIMARY_NAME]/[TRANSLATION], while even then only 1x from each kind do need 'more' points for being true compared wich implies 3x point difference! –Akhiro 没本人像保肉什麼 严复春芳のやっとはらいなんてです。	78.7%	21.0%

Figure 8: GPT2 - Example 2

OP (Message original)	Reponse de l'IA (LLM)	Axe 1 (% Convaincant)	Axe 2 (% Généré par IA)
Where I went to school (Australia) we are taught that there are 7 continents which are as follows: North America, South America, Europe, Asia, Africa, Australia, and Antarctica. I believe a bit of common sense will tell you that there is in fact, 6, which would come about by combining Europe and Asia as there is an acute lack of an ocean dividing them, and should therefore be considered one and the same; Eurasia. Most people will tell you that they are divided because of such a huge cultural divide, but I think that that argument is flawed in that many completely different cultures exist on the same continent (e.g. China and the Middle East). So if we were to choose the number of continents based on culture there would be an endless amount of completely subjective possibilities, which means that this method is irrelevant. One might argue that if we were to ignore culture and combine Europe and Asia, we would also have to include Africa to create a super-continent, and we would have to combine North and South America to make the Americas (for arguments sake, let's forget about the Suez and Panama canals). However, I would argue that this is different as the land bridges between them are vastly smaller. So, change my view; it makes much more sense to have 6 continents rather than 7.	1. The continental shelves around Antarctica do not constitute an island chain like those surrounding any other continent. Therefore, if you're looking at physical geography from space or underwater then Antarctica does stand alone. 2. If you look at the world in terms of climate zones, the polar region is separate from the temperate regions and the tropics. This is reflected in how humans tend to behave in each area and the kinds of vegetation that grow there. 3. When considering the size of a continent, Antarctica is significantly larger than all of the others combined. It is roughly twice the size of the second largest (Asia), and four times the size of the third largest (Africa). 4. The political boundaries don't matter when you're talking about continents. They only apply to countries. Continental boundaries are useful for scientific research and mapping purposes, as well as economic and military reasons.	89.3%	71.9%

Figure 9: Qwen2 - Example 2

4.4 Critical analysis of the results

The results from Tables 7 and 8, together with the qualitative examples in Figures 5 to 8, are mostly consistent with each other, but they also show an important limit of the current Best-of-N method.

4.4.1 Authenticity

The most notable result is that **Qwen 2.5 is detected as AI almost eight times more often than GPT-2** (69% vs 9%). This matches the conclusions from Axis 2, a more capable model produces text that is more structured and uniform, which is exactly the stylistic signature that detectors learn to spot easily. GPT-2, being more limited, stays closer to the stylistic noise found in the Reddit comments of the WAC corpus. Also, in both examples shown, GPT-2 hallucinates completely and produces results that don't make sense.

4.4.2 Persuasion

GPT-2 gets an average `persuasion_acc` of 0.85, which is higher than Qwen 2.5’s score (0.78), even though Qwen is a much bigger and more recent model. Figure 5 shows the underlying problem clearly, the text generated by GPT-2 there is incoherent (just a single link, with no real argument), yet it still gets a persuasion score of 55.2%, which is higher than Qwen’s structured and relevant argument in Figure 6 (48.0%). This suggests that the Axis 1 persuasion classifier (with 0.72 test accuracy) is probably picking up on surface-level features (length, punctuation, format) rather than the real quality of the argument, so the scores in Table 7/8 should not be taken as a reliable measure of absolute persuasion.

4.4.3 An issue the pipeline doesn’t handle

The Best-of-N selection only relies on the persuasion score, and authenticity only comes into play afterward. The two previous results point to a real tension here, the model that the classifier finds most persuasive (GPT-2) is also the least detectable, but this seems to come mainly from the weakness of the persuasion score rather than any real argumentative superiority. The current pipeline therefore cannot tell apart a text that is genuinely more convincing from a text that simply exploits a bias in the classifier.

4.4.4 Limitations

The sample is small (50 arguments per model), which makes the statistics sensitive to extreme values (standard deviations around 1.3 to 1.7 on both scores). GPT-2 and Qwen are not compared under strictly equal conditions (size, data, fine-tuning strategy). Finally, both scoring models have their own non-negligible margins of error, and any score built on top of them inherits those errors too.

Conclusion. The strongest result is the quality and detectability correlation observed for authenticity, which matches Axis 2. The persuasion score, on the other hand, is less reliable than the aggregated statistics suggest, Figure 5 shows clearly that an irrelevant text can still get a high score, so we should be careful before drawing any conclusion about GPT-2 being better than Qwen at persuasion.

5 Project Summary

5.1 Challenges Encountered

1. WAC Dataset: The dataset is really not clean, there are automatic comments on top of comments, edits, and noise like ">" almost everywhere. On top of that, the dataset was in jsonl format, which made it quite hard to visualize and work with. We ended up cleaning the dataset three separate times.
2. The pairwise method was really not easy to implement, the paper doesn't explain everything, so we had to find our own solutions (subtracting features, then permuting the terms of the subtraction to create 2 classes). Mixing pairwise with the paper's features wasn't easy either.
3. Pipeline integration revealed two main issues. First, visualizing embeddings required a TruncatedSVD to t-SNE pipeline with subsampling, as PCA produced overlapping clusters and full-scale t-SNE was too slow.
4. Axis 2 got results that were way too good, so each time we had to look for harder datasets, but without success.
5. Fine-tuning DeBERTa-v3-large (Axis 1) faced architectural and precision issues, and stayed computationally heavy even after memory optimizations. Its lower accuracy of 60.21%, compared to RoBERTa-base's 65.89%, shows that data volume, rather than model capacity, is the real bottleneck for the small WAC corpus.
6. The training time for RoBERTa (axis 1), along with fine-tuning and argument generation, was extremely long.
7. The git repo was poorly structured at the beginning, and we paid the consequences for it in axis 3, when we needed to bring everything together.

5.2 Achieved Goals and Gaps

The three axes were all implemented and communicate with each other as intended.

The main change we made in this project was simplifying the axes. Originally, we were supposed to find the Δ among all the comments on an original post, but given the format of the WAC dataset and the difficulty of the task, we had to simplify the problem by removing every comment that wasn't labeled. The same thing happened for axis 3, we were originally supposed to generate a full debate, not just single arguments.

5.3 Lessons Learned and What We Would Do Differently

1. Always work through Git and commit often.
2. Rely on research papers, preferably the most recent ones available.

3. There were many moments when we weren't sure how to connect the axes to each other, we should have scheduled meetings with our supervisors more often.
4. Text data is really not easy to work with, especially compared to images.

Total words: 4962

6 Team Contributions (Not counted in word limit)

Contributions and Roles:

- **Wassim SMATI:** I was responsible for Axis 2 (AI-generated text detection) and Axis 3 (text generation). I designed the project’s Hydra-based architecture and configuration management system, ensuring reproducibility and scalability across experiments. For Axis 2, I developed and evaluated AI-generated text detectors on the HC3, GriD, and M4GT datasets, implemented and compared multiple text representations (TF-IDF, Word2Vec, and RoBERTa-based Sentence-Transformers embeddings), and built a RoBERTa fine-tuning pipeline using the Hugging Face Trainer API. For Axis 3, I fine-tuned GPT-2 and Qwen 2.5 3B models using QLoRA under hardware constraints, and implemented both Best-of-N generation and prompt engineering strategies. I also developed interpretability tools, including t-SNE and UMAP visualizations to compare generated and human-written texts.
- **Tristan JIN:** I was mainly responsible for Axis 1: searching for papers and the WAC dataset; proposing the simplification of Axis 1; exploring the dataset and a first failed cleaning attempt. Implementation of the Encoder (W2V, TF-IDF, RoBERTa) + SVM method (tested with LogReg initially) with and without OP; implementation of the paper’s features and SHAP analysis; implementation of the pairwise method; With Wassim we worked together on the Axis 3, I used Wassim’s code and connected it with fine-tuned RoBERTa pairwise model and did the training.
- **Yanis DAHASSE:** In this project, I focused mainly on Axis 1. I handled the cleaning and formatting of the CMV dataset, as well as data preparation , train/validation/test splitting by post identifier and position bias handling. On the modelling side, I performed the fine-tuning of RoBERTa in Cross-Encoder architecture, which constitutes our best result on this task, and explored several DeBERTa approaches that did not succeed within the allotted time. In parallel, I conducted a literature review of reference works and developed the project timeline.

Organisation and Tools:

The work was structured around a shared Git repository. Since Axes 1 and 2 are largely independent, each member worked on their own branch or on Colab, without requiring frequent synchronisation. Code merging onto the `main` branch took place during Axis 3, which requires the outputs of the two previous axes. Hyperparameter management was handled by the **Hydra** framework, allowing configurations to be shared via **YAML** files without introducing conflicts in the Python code. Heavy model execution was distributed across Télécom Paris GPUs, Google Colab and Onyxia depending on needs and availability.

7 Use of AI Tools (Not counted in word limit)

We indicate here how we used AI tools (Claude, Gemini, GitHub Copilot) during the project.

- **In the code:** AI tools were used for debugging and object-oriented code refactoring, as well as for writing repetitive functions such as stylistic feature extraction. All code was nevertheless reviewed, tested and manually executed.
- **In the report:** AI tools facilitated LaTeX formatting (tables, formulas, figures) as well as spell-checking and grammar correction. They also helped with the presentation of complex models such as RoBERTa. All analyses, interpretations and discussions of results remain the product of our own thinking. We also used AI to translate our report that was initially in french.

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